THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

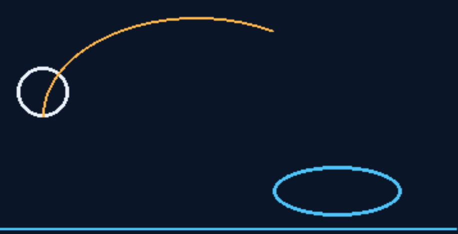
PHASE 179

THE GILLIGAN

'LITTLE BUDDY' THROWABLE LIFE FLOAT

A baseball you can throw far

wet, it fires, pierces, inflates



the capsize box needs nobody

SYPHON CHARGE · WATER SWITCH · COLLAR FLOAT

FLOTATION THAT CANNOT REACH IS NOT THERE

Rescue system — v1.0 in force

GP-IM-179

Revision v1.0 — 23 August 2026

180 of 290

VETTED BATCH 21 — PASS · DEPLOYMENT TESTS OWED

PRIMARY AUTHOR & INVENTOR: ARCHER QUINN — AEROSPACE DESIGN ENGINEER, NPHD

CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 179

PHASE 179: THE GILLIGAN — 'LITTLE BUDDY' THROWABLE SELF-INFLATING LIFE FLOAT — STATUS: CURRENT — PASS; DEPLOYMENT TESTING OWED (VET BATCH 21). Flotation that cannot reach is flotation that is not there. You cannot throw a life ring very far — heavy, draggy, lands where it wants — but a baseball you can throw a long way, accurately: the most practiced throw in human hands. Phase 179 packs the rescue into the ball: a baseball-sized package holding a soda-syphon-class CO_2 charge, a folded noodle-class collar float, and a mechanical/electrical firing pin with a water switch. Dry, it is a ball; in the water the switch fires, the pin pierces the bottle, the float inflates and unwraps. The float's secret is his field data: a pool noodle can hold a hundred kilos at surface level — he knows, he weighs 110 — because a human body in water needs only a few kilograms of buoyancy to keep a face above the surface. The doctrine: single casualty, throw it past the casualty, never at the head; mass casualty, dump the box — dozens of balls self-activate into a field of floats. And the capsize box, the one that needs nobody: floating lid, boat capsizes, lid comes off, the balls float free and activate on contact — the deployment mechanism is the accident itself. The vet passed it; deployment testing is owed, in §9.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-179, v1.0 — 23 August 2026. The one hundred and eightieth manual of the program — 180 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the float law as seated (contents line, the bodies — the problem, the ball, the throw and the dump, the capsize box — the audit and provenance, verbatim) — §2 why the ball is the form factor (graded) — §3 the system in the ball — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 179: **The Gilligan — 'Little Buddy' Throwable Self-Inflating Life Float.** A baseball-sized ball packing a soda-syphon-class CO_2 charge, a folded noodle-class collar float, a mechanical/electrical firing pin, and a water-activated switch: thrown like a baseball (far beyond any life ring's range), it self-inflates on splashdown into a float that keeps even a 110kg casualty's airway clear (bodies are near-neutral; the float buys the airway). Mass deployment by box-dump in a sinking; small boats carry a floating-lid capsize box that deploys itself with no crew action. For cruise ships, planetary shore parties, and naval use.'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Elimination of Flotation Delivery Failure — the Float Reaches the Person, Every Time, Even With No One Left to Throw It *[WORDING KILLED 15 AUGUST 2026 — author's order 'kill anything that says absolute or 100 percent'; original: '100%']*'

Maritime Safety Baseline: Throwable & Self-Deploying Personal Flotation at Scale

Live instruction, 30 July 2026. Sixth invention of the sitting. Named by Archer: The Gilligan — Little Buddy life float. Physics audit appended per file doctrine.

THE PROBLEM — FLOTATION THAT CANNOT REACH IS FLOTATION THAT IS NOT THERE (VERBATIM)

- **You cannot throw a pool noodle or a life ring very far.** A ring is heavy, draggy, and lands where it wants — effective accurate range is a few boat lengths, and in swell a miss by two meters is a drowning. Man-overboard and sinking response is fundamentally a distance-and-numbers problem: get buoyancy TO the person, fast, and to MANY people at once.
- **A baseball you can throw a long way** — and accurately. It is the most practiced throw in human hands. The form factor of the rescue device should match the arm that throws it.

THE BALL (VERBATIM)

- **The package:** baseball-sized ball containing a soda-syphon-bottle-sized CO_2 charge, the float device wrapped/folded inside, a mechanical/electrical firing pin actuator, and a water-activated switch. Air is amazing, but all gas is amazing — the tiny syphon bottle lets out a massive amount of gas relative to its size.
- **The action:** dry, it is a ball. In the water, the switch fires the pin, the pin pierces the bottle, the float inflates and unwraps — a noodle-class collar float where the ball splashed down, seconds after contact.
- **The float's secret, in Archer's field data:** a pool noodle can hold a hundred kilos at surface level — he knows; he weighs 110. The audit's mechanism: a human body is nearly neutrally buoyant in water, so two to four kilograms of float buoyancy is all it takes to keep a large man's airway above the

surface. The float never lifts the person; it buys the airway. That is why a ball-sized package is enough package.

THE THROW & THE DUMP (VERBATIM)

- **Single casualty:** throw it like a baseball — past the casualty, never at the head (rescue-throw doctrine): it lands within arm's reach and becomes a float before they swim a stroke.
- **Mass casualty:** if a ship is sinking, you just dump a box overboard. Dozens of balls hit the water and self-activate — a field of floats for a crowd that no crew could ever throw rings to fast enough.
- **Rails and stations:** cruise ships carry boxes at rails and muster points; the fleet's launches and shore boats carry them as standard kit; planetary shore parties pack them beside the Phase 165 gauntlets — alien water drowns exactly like Earth water.

THE CAPSIZE BOX — THE ONE THAT NEEDS NOBODY (VERBATIM)

- **The small-boat version:** a box with a floating lid. Boat capsizes, the lid comes off, the balls float free and activate on contact — everyone still alive gets a float with no human hand involved at all.
- **Why it is the most important variant:** the capsize is the accident where nobody has time to throw anything. The deployment mechanism is the accident itself — water arrives, floats deploy. This is the hydrostatic-release logic of ship liferaft canisters, reduced to a lid that floats.

PHYSICS NOTE — THE AUDIT (KIMI: AFFIRMED — EVERY GUT IS PROVEN; THE INVENTION IS THE PACKAGE AND THE DOCTRINE) (VERBATIM)

Every mechanism inside the ball is decades-proven hardware: automatic life jackets worldwide inflate on exactly this chain — water-activated trigger (dissolving bobbin or water-sensor circuit), firing pin, CO2 cartridge, buoyancy chamber — and 8-33 grams of CO2 reliably inflates 150-newton-class chambers, so a syphon-bottle charge against a noodle-class collar has gas to spare. The throwability claim is geometry and drag: a life ring's flat profile fights the air and the shoulder, while a baseball is the optimized human projectile — several times the accurate range, with the throw already trained into most arms on Earth. The buoyancy claim is the one most people get wrong and Archer gets right through experience: bodies are near-neutral in water (lungs full, a human floats with only the airway barely clear), so rescue flotation works by topping up a few kilograms, not by lifting body weight — a noodle-class float genuinely serves a 110kg man, and lifeguard rescue tubes of similar buoyancy serve several casualties at once. The capsize box has the oldest provenance of all: ship liferafts have deployed on hydrostatic release for generations — the sea triggers the rescue, no crew required; the floating lid is the same principle with zero moving parts. Flags per doctrine: inflation gas density drops in very cold water (the PFD industry already sizes for it — the collar spec follows suit); the collar must self-present and resist tangling as it unwraps, so the fold pattern is a controlled spec; hi-viz color and a strobe option are standard fit; and the throw doctrine is written into the package — aim past the casualty, because a 150-gram ball arriving at speed is a hazard to a panicked face. His 'everyone still alive survives' stands in the Origin as doctrine's intent; the body text carries the calibrated version — instant flotation at scale removes the largest single killer after cold shock, and the capsize box removes the need for anyone to be heroic first.

ORIGIN OF THOUGHT — PHASE 179 (VERBATIM)

Archer, 30 July 2026: "lifesaver baseball float for all cruise ships and planetary visits and the nave. ok so unknown to most people a pool noodle can hold a hundred kilos at surface level, i know i weigh 110. air is amazing, but all gas amazing, co2 tiny soda syphon bottle size lets out a massive amount of pressure, so that size, in a ball that has the float device attached to it wrapped/folded up inside. the actuator is a mechanical/electrical firing pin. the switch is the same as any water activated switch. you cannot throw a pool noodle or a life ring very far, but a baseball you can throw a long way, if a ship is sinking you just dump a box overboard, the smaller boats can have them in a box that has a floating lid, boat capsizes lid comes off balls activate everyone still alive survives. The Gilligan. Little Buddy life float"

Why it matters, in plain English: drowning gives you minutes and the sea gives you crowds, and every float ever invented before this one had to be carried to the casualty by someone brave and close. A baseball carries itself — sixty meters of trained human arm, a box tipped over a rail, or a lid that floats off a capsized hull while everyone is still gasping. The Skipper would have approved: nobody gets left without a Little Buddy.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE FORM FACTOR, AFFIRMED: rescue range is set by aerodynamics and shoulder strength, and the baseball is the optimized thrown object of the species. Matching the rescue device to the most practiced throw in human hands is human-factors engineering at its cleanest.

THE BUOYANCY MATH, AFFIRMED: a body in water is nearly neutrally buoyant — the float owes only the few kilograms that keep a face above the surface. A noodle-class collar delivers that with margin; his hundred-kilos field datum is the honest upper bound, and the mechanism note is seated with it.

THE WATER-ACTIVATED CHAIN, AFFIRMED AS PROVEN PARTS: water switches, firing pins, syphon charges, and self-inflating collars are each old hardware — every gut is proven; the invention is the package and the doctrine.

THE CAPSIZE BOX IS THE IMPORTANT VARIANT (the audit's emphasis, his design): every other rescue device needs someone present, awake, and able to throw. The capsize is precisely the accident where nobody has time — so the box makes the accident itself pull the trigger.

§3 — THE SYSTEM IN THE BALL

- THE BALL: baseball-sized — CO_2 syphon charge, folded collar float, firing pin, water switch.
- THE ACTION: dry it is a ball; wet it fires, pierces, inflates, unwraps — a noodle-class collar.
- THE THROW: past the casualty, never at the head — lands within arm's reach, inflates before the reach ends.
- THE DUMP: mass casualty — the box overboard; dozens self-activate into a field of floats.
- THE STATIONS: boxes at cruise-ship rails and muster points; launches, shore boats, shore parties carry them standard.
- THE CAPSIZE BOX: floating lid — the capsize itself deploys the balls; the variant that needs nobody.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE REACH DOCTRINE: flotation that cannot reach is flotation that is not there — range is the spec.
- THE PRACTICED-THROW DOCTRINE: match the device to the skill the species already has.
- THE PAST-NOT-AT DOCTRINE: rescue-throw law — past the casualty, never at the head.
- THE NOBODY-REQUIRED DOCTRINE: the worst accident gets the device that deploys itself.

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT), VET BATCH 21 (16 August 2026 — phases 176-180, cross-checked in sitting 316), SEATED. The grade, verbatim from the batch: '179 → PASS / deployment testing.' The deployment program — throw accuracy, activation reliability, inflation time, the capsize box — lives in §9.

§6 — BUILD SEQUENCE

- STEP 1 — Build the ball: package, pin, switch, charge, collar — dry handling safe.
- STEP 2 — Tank-test the activation: fire-on-water reliability, inflation time, collar form.
- STEP 3 — Field the throw (the vet's yellow): accuracy and range across untrained hands; past-not-at doctrine verified.
- STEP 4 — Field the box: dump spread, capsize-lid release, mass-activation count.
- STEP 5 — Publish the ball, the box, and the station doctrine — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 178 — the suppression disc (GP-IM-178): the same one-hand delivery doctrine, fire side.
- PHASE 165 — the tether gauntlets (GP-IM-165): the line-thrower's water cousin.
- PHASE 163 — the casualty doctrine (GP-IM-163): triage extends to the water.
- PHASE 167 — the universal container (GP-IM-167): the station boxes ride the closed-loop standard.

§8 — DO-NOT-BUILD

- DO NOT throw at the head — past the casualty, always; the ball is hard and the doctrine is law.
- DO NOT rely on the ring — it cannot reach; the ball is the reach.
- DO NOT arm it dry — the water switch is the trigger; dry, it stays a ball.
- DO NOT skip the capsize box on small boats — the accident that needs it most gives no time to throw.

§9 — OWED

OWED (the vet's yellow): deployment testing — throw accuracy across users, water-activation reliability, inflation time, capsize-box release (§6). The 15 August blanket wording kills are carried inline in §1.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-179, v1.0 — CURRENT — PASS; deployment testing owed (vet batch 21's yellow, carried in §9), seated law, master v2.1 (numerical print order). The one hundred and eightieth manual of the program — 180 of 290. Built 23 August 2026, twelfth of the 20-manual 'ok another 20 lets keep this train rollin' shift (the author's order, 23 August 2026). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587 and 590): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin'. The phase's own chain, all seated in the master: the contents line, the masthead trio and the four bodies (as seated); the live instruction of 30 July 2026 — sixth invention of the sitting, the Gilligan named by the author; the vet batch 21 grade (PASS; deployment testing owed) in §5 and §9.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the builders'). Next update triggers: the deployment data, or his word, or his word.

END OF MANUAL — PHASE 179